

Sri Lanka Institute of Information Technology



IT2130 – Operating Systems and System Administration Year
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Practical Answer Submission

Practical Sheet No: 09

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Activity 01

1)

```
vboxuser@ubuntu:~$ pwd
/home/vboxuser
```

Absolute path name: /home/vboxuser

2)

```
vboxuser@ubuntu:~$ cd /
vboxuser@ubuntu:/$ ls -l
total 88
lrwxrwxrwx   1 root root    7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x   2 root root 4096 Feb 26  2024 bin.usr-is-merged
drwxr-xr-x   3 root root 4096 Jan 30 10:43 boot
dr-xr-xr-x   2 root root 4096 Jan 30 10:22 cdrom
drwxr-xr-x  19 root root 4280 Apr  5 04:21 dev
drwxr-xr-x 138 root root 12288 Feb  8 08:58 etc
drwxr-xr-x   3 root root 4096 Jan 30 10:44 home
lrwxrwxrwx   1 root root    7 Apr 22  2024 lib -> usr/lib
lrwxrwxrwx   1 root root    9 Apr 22  2024 lib64 -> usr/lib64
drwxr-xr-x   2 root root 4096 Apr  8  2024 lib.usr-is-merged
drwx-----  2 root root 16384 Jan 30 10:24 lost+found
drwxr-xr-x   2 root root 4096 Aug  5  2025 media
drwxr-xr-x   2 root root 4096 Aug  5  2025 mnt
drwxr-xr-x   2 root root 4096 Aug  5  2025 opt
dr-xr-xr-x 277 root root    0 Apr  5 04:21 proc
drwx-----  5 root root 4096 Jan 30 10:44 root
drwxr-xr-x  34 root root  840 Apr  5 04:21 run
lrwxrwxrwx   1 root root    8 Apr 22  2024 sbin -> usr/sbin
drwxr-xr-x   2 root root 4096 Mar 31  2024 sbin.usr-is-merged
drwxr-xr-x  15 root root 4096 Mar 26 08:00 snap
drwxr-xr-x   2 root root 4096 Aug  5  2025 srv
dr-xr-xr-x  13 root root    0 Apr  5 04:21 sys
drwxrwxrwt  17 root root 4096 Apr  5 04:23 tmp
drwxr-xr-x  12 root root 4096 Aug  5  2025 usr
drwxr-xr-x  14 root root 4096 Jan 30 10:44 var
```

How to recognize file/directory:

- d = directory
- - = file
- l = link

Permissions:

- r = read, w = write, x = execute

- Divided into owner, group, others Permission access:
- Most files belong to root
- User has limited permissions (mostly read & execute)

3)

/bin

```
vboxuser@ubuntu:/$ cd /bin
vboxuser@ubuntu:/bin$ ls -l
total 224812
-rwxr-xr-x 1 root root      55744 Apr  5 2024 '['
-rwxr-xr-x 1 root root      18744 Mar 19 2025 aa-enabled
-rwxr-xr-x 1 root root      18744 Mar 19 2025 aa-exec
-rwxr-xr-x 1 root root      18736 Mar 19 2025 aa-features-abi
-rwxr-xr-x 1 root root      22912 Apr  7 2024 aconnect
-rwxr-xr-x 1 root root       1622 Jan  9 16:07 acpidbg
-rwxr-xr-x 1 root root      16422 Feb 18 2025 add-apt-repository
-rwxr-xr-x 1 root root      14720 Jun  5 2025 addpart
lrwxrwxrwx 1 root root         26 Dec  3 15:01 addr2line -> x86_64-linux-gnu-addr2line
-rwxr-xr-x 1 root root     158576 Apr  8 2024 aircan-discover
-rwxr-xr-x 1 root root      47552 Apr  7 2024 alsabat
-rwxr-xr-x 1 root root      85328 Apr  7 2024 alsaloop
-rwxr-xr-x 1 root root      82216 Apr  7 2024 alsamixer
-rwxr-xr-x 1 root root      76160 Apr  7 2024 alsatplg
-rwxr-xr-x 1 root root      31576 Apr  7 2024 alsaucm
-rwxr-xr-x 1 root root      31112 Apr  7 2024 amidi
-rwxr-xr-x 1 root root      63952 Apr  7 2024 amixer
-rwxr-xr-x 1 root root        274 Apr  8 2024 apg
-rwxr-xr-x 1 root root      26784 Apr  8 2024 apgbfm
-rwxr-xr-x 1 root root      84400 Apr  7 2024 aplay
-rwxr-xr-x 1 root root      27016 Apr  7 2024 aplaymidi
-rwxr-xr-x 1 root root       2322 Apr 18 2024 apport-bug
-rwxr-xr-x 1 root root      13625 Jul  8 2025 apport-cli
lrwxrwxrwx 1 root root         10 Jul  8 2025 apport-collect -> apport-bug
-rwxr-xr-x 1 root root       3790 Jul  8 2025 apport-unpack
-rwxr-xr-x 1 root root      14648 Mar 31 2024 appres
-rwxr-xr-x 1 root root     141544 Apr  8 2024 appstreamcli
lrwxrwxrwx 1 root root         6 Apr  8 2024 apropos -> whatis
-rwxr-xr-x 1 root root      18824 Oct 22 2024 ant
```

/home

```
vboxuser@ubuntu:/bin$ cd /home
vboxuser@ubuntu:/home$ ls -l
total 4
drwxr-x--- 16 vboxuser vboxuser 4096 Apr  2 10:29 vboxuser
```


/dev

```
vboxuser@ubuntu:/home$ cd /dev
vboxuser@ubuntu:/dev$ ls -l
total 0
crw-r--r-- 1 root root 10, 235 Apr 5 04:21 autofs
drwxr-xr-x 2 root root 520 Apr 5 04:21 block
drwxr-xr-x 2 root root 80 Apr 5 04:21 bsg
crw----- 1 root root 10, 234 Apr 5 04:21 btrfs-control
drwxr-xr-x 3 root root 60 Apr 5 04:21 bus
lrwxrwxrwx 1 root root 3 Apr 5 04:21 cdrom -> sr0
drwxr-xr-x 2 root root 3760 Apr 5 04:21 char
crw----- 1 root root 5, 1 Apr 5 04:21 console
lrwxrwxrwx 1 root root 11 Apr 5 04:21 core -> /proc/kcore
drwxr-xr-x 6 root root 120 Apr 5 04:21 cpu
crw----- 1 root root 10, 123 Apr 5 04:21 cpu_dma_latency
crw----- 1 root root 10, 203 Apr 5 04:21 cuse
drwxr-xr-x 9 root root 180 Apr 5 04:21 disk
drwxr-xr-x 2 root root 60 Apr 5 04:21 dma_heap
drwxr-xr-x 3 root root 100 Apr 5 04:21 dri
crw----- 1 root root 10, 125 Apr 5 04:21 ecryptfs
crw-rw---- 1 root video 29, 0 Apr 5 04:21 fb0
lrwxrwxrwx 1 root root 13 Apr 5 04:21 fd -> /proc/self/fd
crw-rw-rw- 1 root root 1, 7 Apr 5 04:21 full
crw-rw-rw- 1 root root 10, 229 Apr 5 04:21 fuse
crw----- 1 root root 241, 0 Apr 5 04:21 hidraw0
crw----- 1 root root 10, 228 Apr 5 04:21 hpet
drwxr-xr-x 2 root root 0 Apr 5 04:21 hugepages
crw----- 1 root root 10, 183 Apr 5 04:21 hwrng
crw----- 1 root root 89, 0 Apr 5 04:21 i2c-0
lrwxrwxrwx 1 root root 12 Apr 5 04:21 initctl -> /run/initctl
drwxr-xr-x 4 root root 340 Apr 5 04:21 input
crw-r--r-- 1 root root 1, 11 Apr 5 04:21 kmsg
```

/etc

```
vboxuser@ubuntu:/lib$ cd /etc
vboxuser@ubuntu:/etc$ ls -l
total 1124
-rw-r--r-- 1 root root 3444 Jul 5 2023 adduser.conf
drwxr-xr-x 3 root root 4096 Aug 5 2025 alsa
drwxr-xr-x 2 root root 4096 Feb 8 08:58 alternatives
-rw-r--r-- 1 root root 335 Apr 8 2024 anacrontab
-rw-r--r-- 1 root root 433 Apr 8 2024 apg.conf
drwxr-xr-x 5 root root 4096 Aug 5 2025 apm
drwxr-xr-x 2 root root 4096 Aug 5 2025 apparmor
drwxr-xr-x 9 root root 4096 Jan 30 10:42 apparmor.d
drwxr-xr-x 3 root root 4096 Aug 5 2025 appport
drwxr-xr-x 9 root root 4096 Jan 30 10:24 apt
drwxr-xr-x 3 root root 4096 Jan 30 10:42 avahi
-rw-r--r-- 1 root root 2319 Mar 31 2024 bash.bashrc
-rw-r--r-- 1 root root 45 Jan 24 2020 bash_completion
-rw-r--r-- 1 root root 367 Aug 2 2022 bindresvport.blacklist
drwxr-xr-x 2 root root 4096 Apr 19 2024 binfmt.d
drwxr-xr-x 2 root root 4096 Aug 5 2025 bluetooth
-rw-r----- 1 root root 33 Aug 5 2025 brlapi.key
drwxr-xr-x 7 root root 4096 Aug 5 2025 brltty
-rw-r--r-- 1 root root 30571 Mar 31 2024 brltty.conf
drwxr-xr-x 3 root root 4096 Aug 5 2025 ca-certificates
-rw-r--r-- 1 root root 6288 Aug 5 2025 ca-certificates.conf
drwxr-s-- 2 root dip 4096 Aug 5 2025 chatscripts
drwxr-xr-x 5 root root 4096 Jan 30 10:45 cloud
drwxr-xr-x 2 colord colord 4096 Jan 30 10:44 colord
drwxr-xr-x 2 root root 4096 Jan 30 10:28 console-setup
drwxr-xr-x 2 root root 4096 Aug 5 2025 cracklib
drwx----- 2 root root 4096 Apr 19 2024 credstore
drwx----- 2 root root 4096 Apr 19 2024 credstore.encrypted
```

/lib

```
vboxuser@ubuntu:/dev$ cd /lib
vboxuser@ubuntu:/lib$ ls -l
total 616
drwxr-xr-x  2 root root  4096 Aug  5 2025 app
drwxr-xr-x  2 root root  4096 Aug  5 2025 apparmor
drwxr-xr-x  5 root root  4096 Aug  5 2025 apt
drwxr-xr-x  3 root root 12288 Aug  5 2025 aspell
drwxr-xr-x  2 root root  4096 Feb  8 08:58 bfd-plugins
drwxr-xr-x  2 root root  4096 Jan 30 10:40 binfmt.d
drwxr-xr-x  2 root root  4096 Aug  5 2025 brltty
drwxr-xr-x  2 root root  4096 Aug  5 2025 cloud-init
-rwxr-xr-x  1 root root  1075 Jan 20 2023 cnf-update-db
-rwxr-xr-x  1 root root  3565 Jan 20 2023 command-not-found
drwxr-xr-x  2 root root  4096 Feb  8 08:58 compat-ld
drwxr-xr-x  2 root root  4096 Aug  5 2025 console-setup
lrwxrwxrwx  1 root root    21 Jan 31 2024 cpp -> /etc/alternatives/cpp
drwxr-xr-x 10 root root  4096 Aug  5 2025 cups
drwxr-xr-x  2 root root  4096 Aug  5 2025 dbus-1.0
drwxr-xr-x  3 root root  4096 Aug  5 2025 debug
drwxr-xr-x  3 root root  4096 Aug  5 2025 dhcpcd
drwxr-xr-x  3 root root  4096 Aug  5 2025 dpkg
drwxr-xr-x  2 root root  4096 Aug  5 2025 dracut
drwxr-xr-x  3 root root  4096 Aug  5 2025 emacsen-common
drwxr-xr-x  2 root root  4096 Aug  5 2025 environment.d
drwxr-xr-x  7 root root  4096 Aug  5 2025 evolution-data-server
drwxr-xr-x  2 root root  4096 Aug  5 2025 file
drwxr-xr-x  3 root root  4096 Aug  5 2025 firewallld
drwxr-xr-x 100 root root 36864 Jan 30 10:41 firmware
drwxr-xr-x  3 root root  4096 Feb  8 08:58 gcc
drwxr-xr-x  2 root root  4096 Aug  5 2025 girepository-1.0
drwxr-xr-x  2 root root  4096 Aug  5 2025 gnome-session
drwxr-xr-x  3 root root  4096 Aug  5 2025 gnome-settings-daemon-3.0
```

The ls -l command displays:

- File type (first letter)
- Permissions
- Number of links
- Owner and group
- File size
- Last modified date
- File/directory name

/dev

- Contains device files (hardware devices)
- First letter:
 - C – character device
 - b – block device
 - l – symbolic link
 - d – directory

4)

- cd → go to home directory
- cd . → stay in current directory
- cd .. → go one directory up

Activity 2

1)

```
vboxuser@ubuntu:~$ mkdir IT24104150
vboxuser@ubuntu:~$ cd IT24104150
vboxuser@ubuntu:~/IT24104150$ touch chkpermission
vboxuser@ubuntu:~/IT24104150$ ls -l
total 0
-rw-rw-r-- 1 vboxuser vboxuser 0 Apr  5 12:46 chkpermission
```

2)

```
vboxuser@ubuntu:~/IT24104150$ chmod u=rwx,go= chkpermission
```

3)

```
vboxuser@ubuntu:~/IT24104150$ chmod u=rwx,go= chkpermission
vboxuser@ubuntu:~/IT24104150$ ls -l
total 0
-rwx----- 1 vboxuser vboxuser 0 Apr  5 12:46 chkpermission
```

In octal:


```

vboxuser@ubuntu:~/IT24104150$ touch file2
vboxuser@ubuntu:~/IT24104150$ ls -l
total 0
-rwx----- 1 vboxuser vboxuser 0 Apr  5 12:46 chkpermission
-rw-rw-r-- 1 vboxuser vboxuser 0 Apr  5 12:53 file2
vboxuser@ubuntu:~/IT24104150$ chmod 700 file2
vboxuser@ubuntu:~/IT24104150$ ls -l
total 0
-rwx----- 1 vboxuser vboxuser 0 Apr  5 12:46 chkpermission
-rwx----- 1 vboxuser vboxuser 0 Apr  5 12:53 file2
vboxuser@ubuntu:~/IT24104150$ chmod 755 file2
vboxuser@ubuntu:~/IT24104150$ ls -l
total 0
-rwx----- 1 vboxuser vboxuser 0 Apr  5 12:46 chkpermission
-rwxr-xr-x 1 vboxuser vboxuser 0 Apr  5 12:53 file2
vboxuser@ubuntu:~/IT24104150$

```

4)

```

vboxuser@ubuntu:~/IT24104150$ cd ..
vboxuser@ubuntu:~$ ls -l
total 268
-rwxrwxr-x 1 vboxuser vboxuser 16040 Feb 18 18:00 act1
-rw-rw-r-- 1 vboxuser vboxuser 177 Feb 18 18:00 act1.c
-rwxrwxr-x 1 vboxuser vboxuser 16208 Feb 15 05:11 activity3
-rw-rw-r-- 1 vboxuser vboxuser 791 Feb 15 05:11 activity3.c
-rw-rw-r-- 1 vboxuser vboxuser 1267 Feb 25 13:19 activity5.c
-rwxrwxr-x 1 vboxuser vboxuser 16360 Feb 26 04:02 a.out
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Desktop
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Documents
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Downloads
-rw-rw-r-- 1 vboxuser vboxuser 230 Feb  8 08:42 first.c
-rw-rw-r-- 1 vboxuser vboxuser 1 Feb  8 08:39 first.c.save
-rw-rw-r-- 1 vboxuser vboxuser 30 Mar 26 07:54 fruits.txt
drwxrwxr-x 2 vboxuser vboxuser 4096 Apr  5 12:53 IT24104150
-rwxrwxr-x 1 vboxuser vboxuser 16240 Feb 25 09:16 lab5
-rwxrwxr-x 1 vboxuser vboxuser 16360 Feb 26 04:08 Lab5
-rw-rw-r-- 1 vboxuser vboxuser 1168 Feb 25 10:24 lab5.c
-rw-rw-r-- 1 vboxuser vboxuser 905 Feb 26 04:08 Lab5.c
-rwxrwxr-x 1 vboxuser vboxuser 16216 Mar  1 06:41 lab6
-rw-rw-r-- 1 vboxuser vboxuser 721 Mar  1 06:41 lab6.c
-rwxrwxr-x 1 vboxuser vboxuser 16200 Mar 18 08:06 lab8
-rw-rw-r-- 1 vboxuser vboxuser 453 Mar 18 08:05 lab8.c
-rw-rw-r-- 1 vboxuser vboxuser 0 Mar  1 05:40 lan6.c
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Music
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Public
-rwxrwxr-x 1 vboxuser vboxuser 16312 Mar 26 10:28 q1
-rw-rw-r-- 1 vboxuser vboxuser 910 Mar 26 10:28 q1.c
-rwxrwxr-x 1 vboxuser vboxuser 16416 Mar 26 10:29 q2
-rw-rw-r-- 1 vboxuser vboxuser 1017 Mar 26 10:28 q2.c
-rwxrwxr-x 1 vboxuser vboxuser 16384 Mar 26 10:29 q3

```

```

-rwxrwxr-x 1 vboxuser vboxuser 16384 Mar 26 10:29 q3
-rw-rw-r-- 1 vboxuser vboxuser 2352 Mar 26 10:29 q3.c
-rw-rw-r-- 1 vboxuser vboxuser 2312 Mar 25 07:21 'q3.c  CC'
-rw----- 1 vboxuser vboxuser 19 Mar 25 07:00 q3.n.save
drwx----- 6 vboxuser vboxuser 4096 Feb 8 08:21 snap
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Templates
-rw-rw-r-- 1 vboxuser vboxuser 20 Mar 26 08:07 test.txt
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Videos

```

- Directory permission: rwxr-xr-x

Explanation:

- Owner → read, write, execute
- Group → read, execute
- Others → read, execute

5)

```
vboxuser@ubuntu:~$ chmod u=rwx,go= IT24104150
```

6)

```

vboxuser@ubuntu:~$ chmod u=rwx,go= IT24104150
vboxuser@ubuntu:~$ chmod u=wx IT24104150
vboxuser@ubuntu:~$ ls IT24104150
ls: cannot open directory 'IT24104150': Permission denied

```

7)

```

vboxuser@ubuntu:~$ chmod u=r IT24104150
vboxuser@ubuntu:~$ ls IT24104150
ls: cannot access 'IT24104150/file2': Permission denied
ls: cannot access 'IT24104150/chkpermission': Permission denied
chkpermission file2

```

8)

```

vboxuser@ubuntu:~$ chmod u=rx IT24104150
vboxuser@ubuntu:~$ ls IT24104150
chkpermission file2

```


Activity 3

1)

```
vboxuser@ubuntu:~$ mkdir lab1
vboxuser@ubuntu:~$ cd lab1
vboxuser@ubuntu:~/lab1$ nano idioms
vboxuser@ubuntu:~/lab1$
```

```
GNU nano 7.2 idioms *
Try, Try, Try, One day you can fly

Ideas without action are worthless

What doesn't kill me makes me stronger

Patience is the companion of wisdom

Only those who do nothing makes no mistakes
```

2)

```
vboxuser@ubuntu:~/lab1$ ls -l
total 4
-rw-rw-r-- 1 vboxuser vboxuser 194 Apr  5 13:27 idioms
vboxuser@ubuntu:~/lab1$ ls idioms
idioms
vboxuser@ubuntu:~/lab1$ ls -l
total 4
-rw-rw-r-- 1 vboxuser vboxuser 194 Apr  5 13:27 idioms
vboxuser@ubuntu:~/lab1$ chmod a+rx idioms
vboxuser@ubuntu:~/lab1$ ls -l
total 4
-rwxrwxrwx 1 vboxuser vboxuser 194 Apr  5 13:27 idioms
vboxuser@ubuntu:~/lab1$ chmod 644 idioms
vboxuser@ubuntu:~/lab1$ ls -l
total 4
-rw-r--r-- 1 vboxuser vboxuser 194 Apr  5 13:27 idioms
```

3)

```
vboxuser@ubuntu:~/lab1$ ls -li idioms
1049061 idioms
vboxuser@ubuntu:~/lab1$
```

4)

```
vboxuser@ubuntu:~/lab1$ ls -li idioms
-rw-r--r-- 1 vboxuser vboxuser 194 Apr  5 13:27 idioms
vboxuser@ubuntu:~/lab1$
```

Number of links to file = 1

Activity 4

1)

```
vboxuser@ubuntu:~/lab1$ cd ~  
vboxuser@ubuntu:~$ mkdir linkdemo  
vboxuser@ubuntu:~$ cd linkdemo  
vboxuser@ubuntu:~/linkdemo$ mkdir softln hardln  
vboxuser@ubuntu:~/linkdemo$
```

2)

```
vboxuser@ubuntu:~/linkdemo$ cd hardln  
vboxuser@ubuntu:~/linkdemo/hardln$ ln ~/lab1/idioms hardidioms  
vboxuser@ubuntu:~/linkdemo/hardln$
```

3)

```
vboxuser@ubuntu:~/linkdemo/hardln$ ls -li ~/lab1/idioms hardidioms  
1049061 -rw-r--r-- 2 vboxuser vboxuser 194 Apr  5 13:27 hardidioms  
1049061 -rw-r--r-- 2 vboxuser vboxuser 194 Apr  5 13:27 /home/vboxuser/lab1/idioms  
vboxuser@ubuntu:~/linkdemo/hardln$
```

4)

```
vboxuser@ubuntu:~/linkdemo/hardln$ nano ~/lab1/idioms
vboxuser@ubuntu:~/linkdemo/hardln$ cat hardidioms
X
Try, Try, Try, One day you can fly

Ideas without action are worthless

What doesn't kill me makes me stronger

Patience is the companion of wisdom

Only those who do nothing makes no mistakes

Learning never stops
vboxuser@ubuntu:~/linkdemo/hardln$
```

```
vboxuser@ubuntu:~/linkdemo/hardln$ la -l ~/lab1/idioms hardidioms
-rw-r--r-- 2 vboxuser vboxuser 217 Apr  5 13:44 hardidioms
-rw-r--r-- 2 vboxuser vboxuser 217 Apr  5 13:44 /home/vboxuser/lab1/idioms
vboxuser@ubuntu:~/linkdemo/hardln$ nano hardidioms
vboxuser@ubuntu:~/linkdemo/hardln$ cat ~/lab1/idioms

Try, Try, Try, One day you can fly

Ideas without action are worthless

What doesn't kill me makes me stronger

Patience is the companion of wisdom

Only those who do nothing makes no mistakes

Learning never stops

Practice makes perfect
vboxuser@ubuntu:~/linkdemo/hardln$
```


5)

```
vboxuser@ubuntu:~/linkdemo/hardln$ ln ~/lab1/idioms hardidioms2
vboxuser@ubuntu:~/linkdemo/hardln$ ls -li ~/lab1/idioms hardidioms hardidioms2
1049061 -rw-r--r-- 3 vboxuser vboxuser 240 Apr  5 13:47 hardidioms
1049061 -rw-r--r-- 3 vboxuser vboxuser 240 Apr  5 13:47 hardidioms2
1049061 -rw-r--r-- 3 vboxuser vboxuser 240 Apr  5 13:47 /home/vboxuser/lab1/idioms
vboxuser@ubuntu:~/linkdemo/hardln$ rm hardidioms
vboxuser@ubuntu:~/linkdemo/hardln$ ls -li ~/lab1/idioms hardidioms
ls: cannot access 'hardidioms': No such file or directory
```

6)

```
vboxuser@ubuntu:~/linkdemo/hardln$ cd ~/linkdemo/softln
vboxuser@ubuntu:~/linkdemo/softln$ ln -s ~/lab1/idioms softidioms
vboxuser@ubuntu:~/linkdemo/softln$ ls -l
total 0
lrwxrwxrwx 1 vboxuser vboxuser 26 Apr  5 13:54 softidioms -> /home/vboxuser/lab1/idioms
vboxuser@ubuntu:~/linkdemo/softln$
```

7)

```
vboxuser@ubuntu:~/linkdemo/softln$ ls -li ~/lab1/idioms ~linkdemo/hardln/hardidioms2 softidioms
ls: cannot access '~linkdemo/hardln/hardidioms2': No such file or directory
1049061 -rw-r--r-- 2 vboxuser vboxuser 240 Apr  5 13:47 /home/vboxuser/lab1/idioms
1049462 lrwxrwxrwx 1 vboxuser vboxuser 26 Apr  5 13:54 softidioms -> /home/vboxuser/lab1/idioms
vboxuser@ubuntu:~/linkdemo/softln$
```

8)

```
vboxuser@ubuntu:~/linkdemo/softln$ nano ~/lab1/idioms
vboxuser@ubuntu:~/linkdemo/softln$ cat softidioms

Try, Try, Try, One day you can fly

Ideas without action are worthless

What doesn't kill me makes me stronger

Patience is the companion of wisdom

Only those who do nothing makes no mistakes

Learning never stops

Practice makes perfect

Consistency leads to success
vboxuser@ubuntu:~/linkdemo/softln$ ls -l ~/lab1/idioms softidioms
-rw-r--r-- 2 vboxuser vboxuser 270 Apr  5 13:59 /home/vboxuser/lab1/idioms
lrwxrwxrwx 1 vboxuser vboxuser 26 Apr  5 13:54 softidioms -> /home/vboxuser/lab1/idioms
```

```
vboxuser@ubuntu:~/linkdemo/softln$ nano softidioms
vboxuser@ubuntu:~/linkdemo/softln$ cat ~/lab1/idioms

Try, Try, Try, One day you can fly

Ideas without action are worthless

What doesn't kill me makes me stronger

Patience is the companion of wisdom

Only those who do nothing makes no mistakes

Learning never stops

Practice makes perfect

Consistency leads to success

Keep learning
vboxuser@ubuntu:~/linkdemo/softln$
```

9)

```
vboxuser@ubuntu:~/linkdemo/softln$ cd ~
vboxuser@ubuntu:~$ ln -s ~/lab1/idioms softidioms_new1
vboxuser@ubuntu:~$ ls -li ~/lab1/idioms ~/linkdemo/softln/softidioms softidioms_new1
1049061 -rw-r--r-- 2 vboxuser vboxuser 285 Apr  5 14:02 /home/vboxuser/lab1/idioms
1049462 lrwxrwxrwx 1 vboxuser vboxuser 26 Apr  5 13:54 /home/vboxuser/linkdemo/softln/softidioms -> /home/vboxuser/lab1/idioms
1049463 lrwxrwxrwx 1 vboxuser vboxuser 26 Apr  5 14:07 softidioms_new1 -> /home/vboxuser/lab1/idioms
vboxuser@ubuntu:~$ rm softidioms_new1
vboxuser@ubuntu:~$ ls -li ~/lab1/idioms ~/linkdemo/softln/softidioms
1049061 -rw-r--r-- 2 vboxuser vboxuser 285 Apr  5 14:02 /home/vboxuser/lab1/idioms
1049462 lrwxrwxrwx 1 vboxuser vboxuser 26 Apr  5 13:54 /home/vboxuser/linkdemo/softln/softidioms -> /home/vboxuser/lab1/idioms
vboxuser@ubuntu:~$
```